

1. Data Import:

- Import List of Orders.csv into Power BI.
- Open the list of orders in the Power Query Editor using the Transform Data option.
- Import Order Details.csv and Sales Target.csv into Power Query Editor.

Queries [4]

- List of Orders
- Order Details
- Sales target
- Orders Data

fx = Table.AddColumn(#"Renamed Columns", "Location", each Text.Combine([State], [City]), ", ", type text)

	Order ID	Order Date	CustomerName	State	City	Customer Name Proper Case	Location
1	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Bharat	Gujarat,Ahmedabad
2	B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pearl	Maharashtra,Pune
3	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Jahan	Madhya Pradesh,Bhopal
4	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur	Divsha	Rajasthan,Jaipur
5	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata	Kasheen	West Bengal,Kolkata
6	B-25606	06-04-2018	Hazel	Karnataka	Bangalore	Hazel	Karnataka,Bangalore
7	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Sonakshi	Jammu and Kashmir,Kashmir
8	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai	Aarushi	Tamil Nadu,Chennai
9	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Jitesh	Uttar Pradesh,Lucknow
10	B-25610	09-04-2018	Yogesh	Bihar	Patna	Yogesh	Bihar,Patna
11	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram	Anita	Kerala,Thiruvananthapuram
12	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh	Shrichand	Punjab,Chandigarh
13	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh	Mukesh	Haryana,Chandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla	Vandana	Himachal Pradesh,Simla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok	Bhavna	Sikkim,Gangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa	Kanak	Goa,Goa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima	Sagar	Nagaland,Kohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Manju	Andhra Pradesh,Hyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ramesh	Gujarat,Ahmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune	Sarita	Maharashtra,Pune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Deepak	Madhya Pradesh,Bhopal
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur	Monisha	Rajasthan,Jaipur

List of Orders.csv , Order Details.csv and Sales Target.csv has been imported into power Query

2. Row Limiting & Data Types:

- Restrict the List of Orders table to the first 500 rows.
- Set the Order Date column to the Date data type.
- Change the Amount and Target columns to fixed decimal numbers.

Queries [4] fx = Table.Range(*"Changed Type",0,500)

Order ID	Order Date	CustomerName	State	City
1 B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2 B-25602	01-04-2018	Pearl	Maharashtra	Pune
3 B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4 B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5 B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6 B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7 B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8 B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9 B-25609	09-04-2018	Ittesh	Uttar Pradesh	Lucknow
10 B-25610	09-04-2018	Yogesh	Bihar	Patna
11 B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12 B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13 B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14 B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15 B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16 B-25616	15-04-2018	Kanak	Goa	Goa
17 B-25617	17-04-2018	Sagar	Nagaland	Kohima
18 B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19 B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20 B-25620	20-04-2018	Sarita	Maharashtra	Pune
21 B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22 B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
23 B-25623	22-04-2018	Atharv	West Bengal	Kolkata
24 B-25624	22-04-2018	Vini	Karnataka	Bangalore
25 B-25625	23-04-2018	Pinky	Jammu and Kashmir	Kashmir
26 B-25626	23-04-2018	Bhishm	Maharashtra	Mumbai
27 B-25627	23-04-2018	Hitika	Madhya Pradesh	Indore

Query Settings

PROPERTIES

Name
List of Orders

APPLIED STEPS

Source
Promoted Headers
Changed Type
Kept First 500 Rows
Order Date Data tpe
Customer Name - Proper Case
Renamed Columns
Location- Merged Column

First 500 rows has been restricted in the order List of orders table using Keep rows option.

Queries [4] fx = Table.Range(*"Changed Type",0,500)

Order ID	Order Date	CustomerName	State	City
1 B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
2 B-25602	01-04-2018	Pearl	Maharashtra	Pune
3 B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
4 B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
5 B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
6 B-25606	06-04-2018	Hazel	Karnataka	Bangalore
7 B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
8 B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
9 B-25609	09-04-2018	Ittesh	Uttar Pradesh	Lucknow
10 B-25610	09-04-2018	Yogesh	Bihar	Patna
11 B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
12 B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
13 B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
14 B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
15 B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
16 B-25616	15-04-2018	Kanak	Goa	Goa
17 B-25617	17-04-2018	Sagar	Nagaland	Kohima
18 B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
19 B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
20 B-25620	20-04-2018	Sarita	Maharashtra	Pune
21 B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
22 B-25622	22-04-2018	Monisha	Rajasthan	Jaipur
23 B-25623	22-04-2018	Atharv	West Bengal	Kolkata
24 B-25624	22-04-2018	Vini	Karnataka	Bangalore
25 B-25625	23-04-2018	Pinky	Jammu and Kashmir	Kashmir
26 B-25626	23-04-2018	Bhishm	Maharashtra	Mumbai
27 B-25627	23-04-2018	Hitika	Madhya Pradesh	Indore

Query Settings

PROPERTIES

Name
List of Orders

APPLIED STEPS

Source
Promoted Headers
Changed Type
Kept First 500 Rows
Order Date Data tpe
Customer Name - Proper Case
Renamed Columns
Location- Merged Column

Order Date column data type has been set to Date.

Queries [4]

- List of Orders
- Order Details
- Sales target
- Orders Data

Table.AddColumn(#"Profit Margin - Rounded Off", "Profit Status", each if [Profit Margin] = 0 then "Break-Even" else if [Profit Margin] > 0 then "Profit" else "Loss")

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
1	1,275.00	-1148	7	Furniture	Bookcases		
2	66.00	-12	5	Clothing	Stole		
3	8.00	-2	3	Clothing	Hankerchief		
4	80.00	-56	4	Electronics	Electronic Games		
5	168.00	-111	2	Electronics	Phones		
6	424.00	-272	5	Electronics	Phones		
7	2,617.00	1151	4	Electronics	Phones		
8	561.00	212	3	Clothing	Saree		
9	119.00	-5	8	Clothing	Saree		
10	1,355.00	-60	5	Clothing	Trousers		
11	24.00	-30	1	Furniture	Chairs		
12	193.00	-166	3	Clothing	Saree		
13	180.00	5	3	Clothing	Trousers		
14	116.00	16	4	Clothing	Stole		
15	107.00	36	6	Clothing	Stole		
16	12.00	1	2	Clothing	Hankerchief		
17	38.00	18	1	Clothing	Kurti		
18	65.00	17	2	Clothing	T-shirt		
19	157.00	5	9	Clothing	Saree		
20	75.00	0	7	Clothing	Saree		
21	87.00	4	2	Clothing	Shirt		
22	50.00	15	4	Clothing	Leggings		
23	1,364.00	-1864	5	Furniture	Tables		
24	476.00	0	3	Furniture	Chairs		
25	257.00	23	5	Clothing	Hankerchief		
26	856.00	385	6	Electronics	Printers		
27	485.00	29	4	Electronics	Electronic Games		

Query Settings

PROPERTIES

Name

Order Details

APPLIED STEPS

Source

Promoted Headers

Amount - Fixed Decimal Num...

Added Custom

Changed Type

Profit Margin - Rounded Off

Profit Status - Conditional Co...

Amount Column changed to fixed decimal numbers.

Queries [4]

- List of Orders
- Order Details
- Sales target
- Orders Data

Table.TransformColumnTypes(#"Promoted Headers",{{"Month of Order Date", type date}, {"Category", type text}, {"Target", Currency.Type}})

Month of Order Date	Category	Target
01-04-2018	Furniture	10,400.00
01-05-2018	Furniture	10,500.00
01-06-2018	Furniture	10,600.00
01-07-2018	Furniture	10,800.00
01-08-2018	Furniture	10,900.00
01-09-2018	Furniture	11,000.00
01-10-2018	Furniture	11,100.00
01-11-2018	Furniture	11,300.00
01-12-2018	Furniture	11,400.00
01-01-2019	Furniture	11,500.00
01-02-2019	Furniture	11,600.00
01-03-2019	Furniture	11,800.00
01-04-2018	Clothing	12,000.00
01-05-2018	Clothing	12,000.00
01-06-2018	Clothing	12,000.00
01-07-2018	Clothing	14,000.00
01-08-2018	Clothing	14,000.00
01-09-2018	Clothing	14,000.00
01-10-2018	Clothing	16,000.00
01-11-2018	Clothing	16,000.00
01-12-2018	Clothing	16,000.00
01-01-2019	Clothing	16,000.00
01-02-2019	Clothing	16,000.00
01-03-2019	Clothing	16,000.00
01-04-2018	Electronics	9,000.00
01-05-2018	Electronics	9,000.00
01-06-2018	Electronics	9,000.00
01-07-2018	Electronics	9,000.00

Query Settings

PROPERTIES

Name

Sales target

APPLIED STEPS

Source

Promoted Headers

Target - Fixed Decimal Num...

Target Column changed to fixed decimal numbers.

3. Text Formatting

- Format the CustomerName column to proper case to ensure consistent capitalization.

Order ID	Order Date	CustomerName	State	City	Capitalize Each Word
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Bharat
2	01-04-2018	Pearl	Maharashtra	Pune	Pearl
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Jahan
4	03-04-2018	Divsha	Rajasthan	Jaipur	Divsha
5	05-04-2018	Kasheen	West Bengal	Kolkata	Kasheen
6	06-04-2018	Hazel	Karnataka	Bangalore	Hazel
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Sonakshi
8	08-04-2018	Aarushi	Tamil Nadu	Chennai	Aarushi
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Jitesh
10	09-04-2018	Yogesh	Bihar	Patna	Yogesh
11	11-04-2018	Anita	Kerala	Thiruvananthapuram	Anita
12	12-04-2018	Shrichand	Punjab	Chandigarh	Shrichand
13	12-04-2018	Mukesh	Haryana	Chandigarh	Mukesh
14	13-04-2018	Vandana	Himachal Pradesh	Simla	Vandana
15	15-04-2018	Bhavna	Sikkim	Gangtok	Bhavna
16	15-04-2018	Kanak	Goa	Goa	Kanak
17	17-04-2018	Sagar	Nagaland	Kohima	Sagar
18	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Manju
19	18-04-2018	Ramesh	Gujarat	Ahmedabad	Ramesh
20	20-04-2018	Sarita	Maharashtra	Pune	Sarita
21	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Deepak
22	22-04-2018	Monisha	Rajasthan	Jaipur	Monisha
23	22-04-2018	Atharv	West Bengal	Kolkata	Atharv
24	22-04-2018	Vini	Karnataka	Bangalore	Vini
25	23-04-2018	Pinky	Jammu and Kashmir	Kashmir	Pinky
26	23-04-2018	Bhishm	Maharashtra	Mumbai	Bhishm
27	24-04-2018	Hirika	Madhya Pradesh	Indore	Hirika

CustomerName column has been formatted to proper case to ensure consistent capitalization.

4. Column Creation

- Merge the City and State columns to create a new column named Location in the format: City, State
- Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

Queries [4]

Table.ReorderColumns("#Location- Merged Column",{"Order ID", "Order Date", "CustomerName", "City", "State", "Location", "Customer Name Proper Case"})

	Order Date	CustomerName	City	State	Location	Customer Name Proper Case
1	01-04-2018	Bharat	Ahmedabad	Gujarat	Ahmedabad,Gujarat	Bharat
2	01-04-2018	Pearl	Pune	Maharashtra	Pune,Maharashtra	Pearl
3	03-04-2018	Jahan	Bhopal	Madhya Pradesh	Bhopal,Madhya Pradesh	Jahan
4	03-04-2018	Divsha	Jaipur	Rajasthan	Jaipur,Rajasthan	Divsha
5	05-04-2018	Kasheen	Kolkata	West Bengal	Kolkata,West Bengal	Kasheen
6	06-04-2018	Hazel	Bangalore	Karnataka	Bangalore,Karnataka	Hazel
7	06-04-2018	Sonakshi	Kashmir	Janmu and Kashmir	Kashmir,Janmu and Kashmir	Sonakshi
8	08-04-2018	Aarushi	Chennai	Tamil Nadu	Chennai,Tamil Nadu	Aarushi
9	09-04-2018	Ititesh	Lucknow	Uttar Pradesh	Lucknow,Uttar Pradesh	Ititesh
10	09-04-2018	Yogesh	Patna	Bihar	Patna,Bihar	Yogesh
11	11-04-2018	Anita	Thiruvananthapuram	Kerala	Thiruvananthapuram,Kerala	Anita
12	12-04-2018	Shrichand	Chandigarh	Punjab	Chandigarh,Punjab	Shrichand
13	12-04-2018	Mukesh	Chandigarh	Haryana	Chandigarh,Haryana	Mukesh
14	13-04-2018	Vandana	Simla	Himachal Pradesh	Simla,Himachal Pradesh	Vandana
15	15-04-2018	Bhavna	Gangtok	Sikkim	Gangtok,Sikkim	Bhavna
16	15-04-2018	Kanak	Goa	Goa	Goa,Goa	Kanak
17	17-04-2018	Sagar	Kohima	Nagaland	Kohima,Nagaland	Sagar
18	18-04-2018	Manju	Hyderabad	Andhra Pradesh	Hyderabad,Andhra Pradesh	Manju
19	18-04-2018	Ramesh	Ahmedabad	Gujarat	Ahmedabad,Gujarat	Ramesh
20	20-04-2018	Sarita	Pune	Maharashtra	Pune,Maharashtra	Sarita
21	20-04-2018	Deepak	Bhopal	Madhya Pradesh	Bhopal,Madhya Pradesh	Deepak
22						

Tamil Nadu

Query Settings

PROPERTIES

Name

List of Orders

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Kept First 500 Rows

Order Date Data type

Customer Name - Proper Case

Renamed Columns

Location- Merged Column

Reordered Columns1

Reordered Columns

City and state column has been merged to create a Location column using comma Seperator.

Queries [4]

Table.TransformColumns("#Changed Type",{"Profit Margin", each Number.Round(, 2), type number})

	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin
1	1,275.00	-1148	7	Furniture	Bookcases	-0.9
2	66.00	-12	5	Clothing	Stole	-0.18
3	8.00	-2	3	Clothing	Hankerchief	-0.25
4	80.00	-56	4	Electronics	Electronic Games	-0.7
5	168.00	-111	2	Electronics	Phones	-0.66
6	424.00	-272	5	Electronics	Phones	-0.64
7	2,617.00	1151	4	Electronics	Phones	0.44
8	561.00	212	3	Clothing	Saree	0.38
9	119.00	-5	8	Clothing	Saree	-0.04
10	1,355.00	-60	5	Clothing	Trousers	-0.04
11	24.00	-30	1	Furniture	Chairs	-1.25
12	193.00	-166	3	Clothing	Saree	-0.86
13	180.00	5	3	Clothing	Trousers	0.03
14	116.00	16	4	Clothing	Stole	0.14
15	107.00	36	6	Clothing	Stole	0.34
16	12.00	1	2	Clothing	Hankerchief	0.08
17	38.00	18	1	Clothing	Kurti	0.47
18	65.00	17	2	Clothing	T-shirt	0.26
19	157.00	5	9	Clothing	Saree	0.03
20	75.00	0	7	Clothing	Saree	0
21	87.00	4	2	Clothing	Shirt	0.05
22	50.00	15	4	Clothing	Leggings	0.3
23	1,364.00	-1864	5	Furniture	Tables	-1.37
24	476.00	0	3	Furniture	Chairs	0
25	257.00	23	5	Clothing	Hankerchief	0.09
26	856.00	385	6	Electronics	Printers	0.45
27	485.00	29	4	Electronics	Electronic Games	0.06

Query Settings

PROPERTIES

Name

Order Details

All Properties

APPLIED STEPS

Source

Promoted Headers

Amount - Fixed Decimal Num...

Added Custom

Changed Type

Profit Margin - Rounded Off

Profit Status - Conditional Co...

Created a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

5. Conditional Column

- Add a conditional column named Profit Status based on:

- Profit < 0 → Loss
- Profit = 0 → Break-Even
- Profit > 0 → Profit

The screenshot shows a data table with columns: Amount, Profit, Quantity, Category, Sub-Category, Profit Margin, and Profit Status. The Profit Status column is calculated based on the Profit Margin column using a conditional formula. The table contains 29 rows of data. The Profit Status column values are: Loss, Loss, Loss, Loss, Loss, Loss, Profit, Profit, Loss, Loss, Loss, Loss, Loss, Profit, Profit, Profit, Profit, Profit, Profit, Profit, Profit, Profit, Profit, Profit, Profit, Profit, Profit, Profit, Profit, Profit, Profit.

	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
1	1,275.00	-248	7	Furniture	Bookcases	-0.9	Loss
2	66.00	-12	5	Clothing	Stole	-0.18	Loss
3	8.00	-2	3	Clothing	Hankkerchief	-0.25	Loss
4	80.00	-8	4	Electronics	Electronic Games	-0.7	Loss
5	168.00	-111	2	Electronics	Phones	-0.66	Loss
6	424.00	-272	5	Electronics	Phones	-0.64	Loss
7	2,617.00	1151	4	Electronics	Phones	0.44	Profit
8	561.00	212	2	Clothing	Saree	0.38	Profit
9	115.00	5	8	Clothing	Saree	-0.04	Loss
10	1,355.00	-40	5	Clothing	Trousers	-0.04	Loss
11	24.00	-30	1	Furniture	Chairs	-1.25	Loss
12	193.00	-166	3	Clothing	Saree	-0.86	Loss
13	180.00	5	2	Clothing	Trousers	0.02	Profit
14	116.00	16	4	Clothing	Stole	0.14	Profit
15	107.00	36	6	Clothing	Stole	0.34	Profit
16	12.00	1	2	Clothing	Hankkerchief	0.08	Profit
17	38.00	18	1	Clothing	Kurti	0.47	Profit
18	65.00	17	2	Clothing	T-shirt	0.26	Profit
19	157.00	5	9	Clothing	Saree	0.03	Profit
20	75.00	0	7	Clothing	Saree	0	Break-Even
21	87.00	4	2	Clothing	Shirt	0.05	Profit
22	80.00	13	4	Clothing	Leggings	0.3	Profit
23	1,364.00	-1864	5	Furniture	Tables	-1.37	Loss
24	476.00	0	3	Furniture	Chairs	0	Break-Even
25	257.00	23	5	Clothing	Hankkerchief	0.09	Profit
26	856.00	385	6	Electronics	Printers	0.45	Profit
27	485.00	29	4	Electronics	Electronic Games	0.06	Profit

Profit status has been created using conditional column option based on below criteria

- Profit < 0 → Loss
- Profit = 0 → Break-Even
- Profit > 0 → Profit

6. Merging Data (Joins)

- Merge the List of Orders and Order Details tables using Order ID.
- Name the merged output table as Orders Data.

Queries [4] Table.ExpandTableColumn(Source, "Order Details", ("Amount", "Profit", "Quantity", "Category", "Sub-Category", "Profit Margin", "Profit"))

Query Settings

PROPERTIES

Name

Order Data

APPLIED STEPS

Source

Expanded Order Details

Order ID	Order Date	Customer Name	State	City	Location	Customer
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	Bharat
2	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	Bharat
3	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	Bharat
4	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad,Gujarat	Bharat
5	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra	Pearl
6	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra	Pearl
7	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra	Pearl
8	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra	Pearl
9	01-04-2018	Pearl	Maharashtra	Pune	Pune,Maharashtra	Pearl
10	01-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Jahan
11	01-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Jahan
12	01-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Jahan
13	01-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Jahan
14	01-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Jahan
15	01-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Jahan
16	01-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Jahan
17	01-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal,Madhya Pradesh	Jahan
18	01-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan	Divsha
19	01-04-2018	Divsha	Rajasthan	Jaipur	Jaipur,Rajasthan	Divsha
20	01-04-2018	Kasheen	West Bengal	Kolkata	Kolkata,West Bengal	Kasheen
21	01-04-2018	Hazel	Karnataka	Bangalore	Bangalore,Karnataka	Hazel
22	01-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir,Jammu and Kashmir	Sonakshi
23	01-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Aarushi
24	01-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Aarushi
25	01-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Aarushi
26	01-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai,Tamil Nadu	Aarushi
27	01-04-2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow,Uttar Pradesh	Jitesh

The **Orders** and **Order Details** tables were merged using the **Merge Queries** feature with a **Left Join**, using **Order ID** as the common key. Merged table was renamed to **Orders Data**.

7. Handling Missing Data & Duplicate Data

- Identify missing values across all tables.
- Define and apply appropriate strategies such as:
 - Replacing missing values
 - Removing records
 - Retaining nulls with justification
- Identify duplicate rows and define a suitable handling strategy based on business logic.

Queries [4] Table.AddColumn("#Renamed Columns", "Location", each Text.Combine([State], [City]), ", ", type text)

Query Settings

PROPERTIES

Name

List of Orders

APPLIED STEPS

Source

Promoted Headers

Changed Type

Kept First 500 Rows

Order Date Data type

Customer Name - Proper Case

Renamed Columns

Location- Merged Column

Order ID	Order Date	Customer Name	State	City	Location	Customer Name Proper Case	Location- Merged Column
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Ahmedabad	Bharat	Gujarat
2	01-04-2018	Pearl	Maharashtra	Pune	Pune	Pearl	Maharashtra
3	01-04-2018	Jahan	Madhya Pradesh	Bhopal	Bhopal	Jahan	Madhya Pradesh
4	01-04-2018	Divsha	Rajasthan	Jaipur	Jaipur	Divsha	Rajasthan
5	01-04-2018	Kasheen	West Bengal	Kolkata	Kolkata	Kasheen	West Bengal
6	01-04-2018	Hazel	Karnataka	Bangalore	Bangalore	Hazel	Karnataka
7	01-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Kashmir	Sonakshi	Jammu and Kashmir
8	01-04-2018	Aarushi	Tamil Nadu	Chennai	Chennai	Aarushi	Tamil Nadu
9	01-04-2018	Jitesh	Uttar Pradesh	Lucknow	Lucknow	Jitesh	Uttar Pradesh
10	01-04-2018	Yogesh	Bihar	Patna	Patna	Yogesh	Bihar
11	01-04-2018	Anita	Kerala	Thiruvananthapuram	Thiruvananthapuram	Anita	Kerala
12	01-04-2018	Shrichand	Punjab	Chandigarh	Chandigarh	Shrichand	Punjab
13	01-04-2018	Mukesh	Haryana	Chandigarh	Chandigarh	Mukesh	Haryana
14	01-04-2018	Vandana	Himachal Pradesh	Simla	Simla	Vandana	Himachal Pradesh

Queries [4] fx = Table.AddColumn(#"Profit Margin - Rounded Off", "Profit Status", each if [Profit Margin] = 0 then "Break-Even" else if [Profit Margin] > 0 then "Profit" else "Loss")

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status
1 B-25601	1,275.00	-1148	7	Furniture	Bookcases	0%	Break-Even
2 B-25601	66.00	-12	5	Clothing	Stole	0%	Break-Even
3 B-25601	8.00	-2	3	Clothing	Hankerchief	0%	Break-Even
4 B-25601	80.00	-56	4	Electronics	Electronic Games	0%	Break-Even
5 B-25602	168.00	-111	2	Electronics	Phones	0%	Break-Even
6 B-25602	424.00	-272	5	Electronics	Phones	0%	Break-Even
7 B-25602	2,617.00	1151	4	Electronics	Phones	100%	Profit
8 B-25602	561.00	212	3	Clothing	Saree	100%	Profit
9 B-25602	119.00	-5	8	Clothing	Saree	0%	Break-Even
10 B-25603	1,355.00	-60	5	Clothing	Trousers	0%	Break-Even
11 B-25603	24.00	-30	1	Furniture	Chairs	0%	Break-Even

Query Settings

PROPERTIES

Name: Order Details

APPLIED STEPS

Source: Order Details

Amount - Fixed Decimal Num...

Added Custom

Changed Type

Profit Margin - Rounded Off

Profit Status - Conditional Co...

Queries [4] fx = Table.TransformColumnTypes(#"Promoted Headers",{{"Month of Order Date", type date}, {"Category", type text}, {"Target", Currency.Type}})

Month of Order Date	Category	Target
01-04-2018	Furniture	10,400.00
01-05-2018	Furniture	10,500.00
01-06-2018	Furniture	10,600.00
01-07-2018	Furniture	10,800.00

Queries [4] fx = Table.ExpandTableColumn(Source, "Order Details", {"Amount", "Profit", "Quantity", "Category", "Sub-Category", "Profit Margin", "Profit Status"})

Order ID	Order Date	Customer Name	State	City	Customer Name Proper Case	Location
1 B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Bharat	Gujarat
2 B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Bharat	Gujarat
3 B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Bharat	Gujarat
4 B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad	Bharat	Gujarat
5 B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pearl	Maharashtra
6 B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pearl	Maharashtra
7 B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pearl	Maharashtra
8 B-25602	01-04-2018	Pearl	Maharashtra	Pune	Pearl	Maharashtra

Query Settings

PROPERTIES

Name: Orders Data

APPLIED STEPS

Source: Expanded Order Details

I verified the data for missing values using **Power Query** → **View** → **Column Quality**, and no missing or null values were identified across any of the tables. I also manually filtered each relevant column for **(Blank)** values, and no blank records were found.

If any Null or missing values found we can fix them by statistical method based on data type though Transform→Statistics→(Mean,Median) for numeric data. Home→Group by for Categorical data. After determining the appropriate statistical replacement value, Home → Replace Values is used to replace the null or missing values.

Removing records→ Remove rows containing missing values only when the number of affected records is minimal and their removal is unlikely to have a significant impact on the overall analysis.

Retaining nulls with justification: Retain null values when they have a meaningful business interpretation (for example, "Not Applicable," "Unknown," or data not yet collected), or when replacing or removing them could Affect the analysis.

Queries [4] Table.Distinct("#Location- Merged Column", ("Order ID"))

Order ID	Order Date	CustomerName	State	City	Customer Name Proper Case	Loca
1	01-04-2018	Bharat	Gujarat	Ahmedabad	Bharat	Gujar
2	01-04-2018	Pearl	Maharashtra	Pune	Pearl	Maha
3	03-04-2018	Jahan	Madhya Pradesh	Bhopal	Jahan	Madh
4	03-04-2018	Divsha	Rajasthan	Jaipur	Divsha	Rajas
5	05-04-2018	Kasheen	West Bengal	Kolkata	Kasheen	West
6	06-04-2018	Hazel	Karnataka	Bangalore	Hazel	Karna
7	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir	Sonakshi	Jamm
8	08-04-2018	Aarushi	Tamil Nadu	Chennai	Aarushi	Tamil
9	09-04-2018	Jitesh	Uttar Pradesh	Lucknow	Jitesh	Uttar
10	09-04-2018	Yogesh	Bihar	Patna	Yogesh	Bihar
11	11-04-2018	Anita	Kerala	Thiruvananthapuram	Anita	Keral
12	12-04-2018	Shrichand	Punjab	Chandigarh	Shrichand	Punjab
13	12-04-2018	Mukesh	Haryana	Chandigarh	Mukesh	Harya

Query Settings

PROPERTIES

Name: List of Orders

APPLIED STEPS

- Source
- Promoted Headers
- Changed Type
- Kept First 500 Rows
- Order Date Data tpe
- Customer Name - Proper Case
- Renamed Columns
- Location- Merged Column
- Removed Duplicates

Duplicate rows Has been deleted through Home—>Remove Rows—>Remove Duplicates

8. Sorting and Filtering Data

- Sort records in the Orders Data table by Order Date in descending order.
- Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

Queries [4] Table.Sort("#Expanded Order Details", ("Order Date", Order.Descending))

Order ID	Order Date	CustomerName	State	City	Customer Name Proper Case	Loca
1	31-03-2019	Hitika	Madhya Pradesh	Indore	Hitika	Madh
2	31-03-2019	Hitika	Madhya Pradesh	Indore	Hitika	Madh
3	31-03-2019	Hitika	Madhya Pradesh	Indore	Hitika	Madh
4	30-03-2019	Bhishm	Maharashtra	Mumbai	Bhishm	Maha
5	30-03-2019	Bhishm	Maharashtra	Mumbai	Bhishm	Maha
6	30-03-2019	Bhishm	Maharashtra	Mumbai	Bhishm	Maha
7	30-03-2019	Bhishm	Maharashtra	Mumbai	Bhishm	Maha
8	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Pinky	Jamm
9	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Pinky	Jamm
10	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Pinky	Jamm
11	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Pinky	Jamm
12	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Pinky	Jamm
13	29-03-2019	Pinky	Jammu and Kashmir	Kashmir	Pinky	Jamm

Query Settings

PROPERTIES

Name: Orders Data

APPLIED STEPS

- Source
- Expanded Order Details
- Order Date - descending Order

Sorted the **Order Date** column in descending order in the **Orders** table.

Order ID	Order Date	Customer Name	State	City	Customer Name Proper Case	Location
1	22-03-2019	Aarushi	Tamil Nadu	Chennai	Aarushi	Tamil
2	22-03-2019	Aarushi	Tamil Nadu	Chennai	Aarushi	Tamil
3	22-03-2019	Aarushi	Tamil Nadu	Chennai	Aarushi	Tamil
4	22-03-2019	Aarushi	Tamil Nadu	Chennai	Aarushi	Tamil
5	22-03-2019	Aarushi	Tamil Nadu	Chennai	Aarushi	Tamil
6	22-03-2019	Aarushi	Tamil Nadu	Chennai	Aarushi	Tamil
7	14-02-2019	Aarushi	Tamil Nadu	Chennai	Aarushi	Tamil
8	14-02-2019	Aarushi	Tamil Nadu	Chennai	Aarushi	Tamil
9	09-02-2019	Kalyani	Tamil Nadu	Chennai	Kalyani	Tamil
10	09-02-2019	Kalyani	Tamil Nadu	Chennai	Kalyani	Tamil
11	09-02-2019	Kalyani	Tamil Nadu	Chennai	Kalyani	Tamil

Filter applied to analyze order from specific State.

9. Grouping and Aggregating Data

- Duplicate the Order Details table and perform the following:
 - Count of Order ID
 - Average Profit by Category
 - Total Amount by Sub-Category
- Duplicate the Sales Target table and aggregate the total Target by Month of order date.

Order ID	Count
1	4
2	5
3	6
4	2
5	2
6	1
7	1
8	4
9	2
10	6
11	1
12	1
13	1
14	2
15	1
16	4
17	1
18	2
19	1
20	1
21	3
22	1
23	4
24	1
25	2

Duplicated the Order Details table and used the Group By feature to count the occurrences of each Order ID.

Queries [5] ✕ ✓ f_x Table.Group(#"Profit Status - Conditional Column", {"Category"}, {"Count", each List.Average([Profit]), type nullable number})

Category	Count
Valid	100%
Error	0%
Empty	0%
1 Furniture	9.456790133
2 Clothing	11.76290832
3 Electronics	34.07142857

Query Settings ✕

PROPERTIES

Name
Order Details

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Amount - Fixed Decimal Num...
- Added Custom
- Changed Type
- Profit Margin - Rounded Off
- Profit Status - Conditional Co...
- Average By Category**

Category wise average has been calculated using Group by Method

Group by: **Category**

Operation: **Average**

Column: **Profit**

Queries [5] ✕ ✓ f_x Table.Group(#"Profit Status - Conditional Column", {"Sub-Category"}, {"Total Amount - Subcategory", each List.Sum([Amount]), type ...})

Sub-Category	Total Amount - Subcategory
Valid	100%
Error	0%
Empty	0%
1 Bookcases	56861
2 Stole	18546
3 Handkerchief	14608
4 Electronic Games	39168
5 Phones	46119
6 Saree	53511
7 Trousers	30039
8 Chairs	34222
9 Kurti	3361
10 T-shirt	7362
11 Shirt	7555
12 Leggings	2106
13 Tables	22614
14 Printers	58252
15 Accessories	21728
16 Furnishings	13484
17 Skirt	1946

Query Settings ✕

PROPERTIES

Name
Order Details - Duplicate

All Properties

APPLIED STEPS

- Source
- Promoted Headers
- Amount - Fixed Decimal Num...
- Added Custom
- Changed Type
- Profit Margin - Rounded Off
- Profit Status - Conditional Co...
- Grouped Rows**

Group By ✕

Specify the column to group by and the desired output.

☒ Basic ☐ Advanced

Sub-Category

New column name: Total Amount - Subcategory

Operation: Sum

Column: Amount

Total Amount by Sub-Category has been calculated using

Home → Group By

- Group by: Sub-Category

- Operation: Sum
- Column: Amount/Sales

Queryes [5]

Table.Group("Target - Fixed Decimal Numbers", {"Month of Order Date"}, ({ "Total Target", each List.Sum([Target]), type nullable number}))

	Month of Order Date	1.2 Total Target
Valid	100%	Valid 100%
Error	0%	Error 0%
Empty	0%	Empty 0%
1	01-04-2018	31400
2	01-05-2018	31500
3	01-06-2018	31600
4	01-07-2018	31800
5	01-08-2018	31900
6	01-09-2018	34000
7	01-10-2018	36100
8	01-11-2018	36300
9	01-12-2018	36400
10	01-01-2019	43500
11	01-02-2019	43600
12	01-03-2019	43800

Query Settings

PROPERTIES

Name

Sales target - Duplicate

APPLIED STEPS

Source

Promoted Headers

Target - Fixed Decimal Num...

Total Target

Duplicated the Sales Target table and the total Target by Month of order date has been calculated.

10. Data Modeling

- Establish a relationship between:
 - List of Orders and Order Details using Order ID
 - Establish a relationship between:
 - Order Details and Sales Target using Category
2. Use Manage Relationships to ensure relationships are active and correctly defined.

New relationship

Select tables and columns that are related.

From table

Order Details

Category	Order ID	Profit Margin	Profit	Profit Status	Quantity	Sub-Categ
Clothing	B-25603	0.08	1	Profit	2	Hankerc
Clothing	B-25608	0.09	23	Profit	5	Hankerc
Clothing	B-25615	0.29	20	Profit	5	Hankerc

To table

List of Orders

City	Customer Na...	CustomerName	Location	Order Date	Order ID	State
Indore	Hitika	Hitika	Madhya Prad...	23 April 2018	B-25627	Madhya
Indore	Ashmi	Ashmi	Madhya Prad...	26 April 2018	B-25637	Madhya
Indore	Yaanvi	Yaanvi	Madhya Prad...	01 May 2018	B-25645	Madhya

Cardinality

Many to one (*:1)

Cross-filter direction

Single

☒ Make this relationship active

☐ Assume referential integrity

☐ Apply security filter in both directions

There's already a relationship between these two columns.

Save Cancel

A relationship between the *List of Orders* and *Order Details* tables has been established using the *Order ID* column as the common key.

New relationship

Select tables and columns that are related.

From table

Order Details

Category	Order ID	Profit Margin	Profit	Profit Status	Quantity	Sub-Category
Clothing	B-25603	0.08	1	Profit	2	Hankerc
Clothing	B-25608	0.09	23	Profit	5	Hankerc
Clothing	B-25615	0.29	20	Profit	5	Hankerc

To table

Sales target

Category	Month of Ord...
Furniture	01 April 2018
Furniture	01 May 2018
Furniture	01 June 2018

Cardinality

Many to many (*:*)

Cross-filter direction

Both

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

 This relationship has cardinality Many-Many. This should only be used if it is expected that neither column (Category and Category) contains unique values, and that the significantly different behavior of Many-Many relationships is understood. [Learn more](#)

A relationship between the *Order Details* and *Sales Target* using *Category* has been established using the *Category* as the common key.

Manage relationships



+ New relationship

⚡ Autodetect

✎ Edit

🗑 Delete

≡ Filter ▾

☐	From: table (column) ↑	Relationship	To: table (column)	Status	
☐	Order Details (Category)	* — ◀ — *	Sales target (Category)	Active	...
☐	Order Details (Order ID)	* — ◀ — 1	List of Orders (Order ID)	Active	...
☐	Order Details (Sub-Category)	* — ◀ — 1	Order Details - Duplicate (Sub...	Active	...
☐	Orders Data (Order Details.Su...	* — ◀ — 1	Order Details - Duplicate (Sub...	Active	...
☐	Orders Data (Order ID)	* — ◀ — 1	List of Orders (Order ID)	Active	...

Close

Used **Manage Relationships** in Power BI to verify that all table relationships were correctly defined and active.